Equations with Brackets & Fractions

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Written by: Mark Curtis
Reviewed by: Dan Finlay
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Equations with brackets & fractions

How do I solve linear equations with brackets?

- If a linear equation involves brackets, **expand** the brackets first
- For example, solve $2(x-3)=10$
  - Expand the brackets

$$2x-6=10$$

- This now has the same form as before
  - Solve by adding 6 then dividing by 2

$$2x=16$$
$$x=\frac{16}{2}$$
$$x=8$$

- **Expanding** brackets **first** will always work, but you can also divide first
  - Dividing **both** sides of $2(x-3)=10$ by 2 gives $(x-3)=5$
    - which solves to $x=8$
  - This method works but can lead to harder **fractions**

How do I solve linear equations with fractions?

- If a linear equation contains **fractions**, **multiply both sides** by the **lowest common denominator**
- For example, solve $\frac{x}{5}+4=\frac{9}{2}$
  - The lowest common denominator of 5 and 2 is 10
  - Multiply **all terms** on both sides by 10
    - Don't forget to do it to the 4

$$10\times\frac{x}{5}+10\times4=10\times\frac{9}{2}$$

- **Simplify** the fractions
  - Imagine the 10 is in the numerator and cancel

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- Solve by subtracting 40 then dividing by 2

$$2x = 5$$
$$x = \frac{5}{2}$$

- Unless the question asks, you can leave the answer like this
  - A **decimal** or **mixed number** would also be **accepted**
- Sometimes equations have the unknown on the denominator
- For example $\frac{4}{x-2} = 3$
  - Multiply both sides of the equation by the denominator

$$\frac{4}{x-2} \times (x-2) = 3(x-2)$$

- Simplify the fractions, and expand any brackets

$$4 = 3(x-2)$$

$$4 = 3x - 6$$

- This now looks like a standard linear equation
  - Solve by adding 6 to both sides, then dividing by 3

$$10 = 3x$$
$$\frac{10}{3} = x$$

Examiner Tips and Tricks

In the exam, always substitute your answer back into the equation to check you got it right!

Worked Example

(a) Solve the equation

$$5(3-4x) + 1 = 26$$

Answer:

First expand the brackets on the left-hand side

$$15 - 20x + 1 = 26$$

Simplify the left-hand side (by adding 15 and 1)

$$16 - 20x = 26$$

Remember 16 - 20x is the same as -20x + 16
Subtract 16 from both sides

Divide both sides by -20 and simplify

$$x = \frac{10}{-20}$$

$$x = -\frac{1}{2}$$

(b) Solve the equation

$$\frac{5x}{4} = \frac{1}{2}$$

Answer:

The lowest common denominator of 4 and 2 is 4
Multiply both sides by 4

$$4 \times \frac{5x}{4} = 4 \times \frac{1}{2}$$

Simplify (cancel) the fractions

$$5x = 2$$

To solve this equation, divide both sides by 5

$$x = \frac{2}{5}$$

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